

# Indices and surds - Set 1

TMUA practice questions by topic. Specification MM1.1, MM1.2, M2.7, M2.11, M2.12. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

1 Simplify  $\frac{(2x^3)^4 \times 8x^{-5}}{4x^2}$ .

A  $32x^9$

B  $16x^7$

C  $x^5$

D  $16x^5$

E  $4x^5$

F  $32x^5$

2 Find the value of  $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$ .

A  $\frac{9}{4}$

B  $\frac{4}{9}$

C  $\frac{9}{4}$

D  $\frac{4}{9}$

E  $\frac{3}{2}$

F  $\frac{2}{3}$

3 Solve  $4^{x+1} = 8^{2x-3}$ .

A  $x = \frac{11}{5}$

B  $x = \frac{4}{9}$

C  $x = \frac{4}{7}$

D  $x = \frac{2}{2}$

E  $x = 2$

F  $x = 4$

- 4 Express  $\frac{5}{3+2\sqrt{5}}$  in the form  $a+b\sqrt{5}$  where  $a$  and  $b$  are rational.
- A  $\frac{15-10\sqrt{5}}{-15+10\sqrt{5}}$   
 B  $\frac{15+10\sqrt{5}}{15-10\sqrt{5}}$   
 C  $\frac{11}{-15+10\sqrt{5}}$   
 D  $\frac{11}{-15+10\sqrt{5}}$   
 E  $\frac{-15+10\sqrt{5}}{-15+10\sqrt{5}}$   
 F  $\frac{-15+10\sqrt{5}}{29}$

- 5 Simplify  $\sqrt{75} - \sqrt{27} + \sqrt{12}$ .
- A  $2\sqrt{3}$   
 B  $6\sqrt{3}$   
 C  $\sqrt{60}$   
 D  $10\sqrt{3}$   
 E  $4\sqrt{3}$   
 F  $5\sqrt{3}$

- 6 Solve  $9^x - 4 \times 3^x + 3 = 0$ .
- A  $x = 1$  only  
 B  $x = 0$  or  $x = 1$   
 C  $x = -1$  or  $x = 0$   
 D  $x = 0$  only  
 E  $x = \log_3 4$   
 F  $x = 1$  or  $x = 3$

- 7 The value of  $\frac{2^{n+2} + 2^n}{2^{n+1}}$  is
- A  $2^n$   
 B  $5$   
 C  $\frac{3}{2}$   
 D  $3$   
 E  $\frac{5}{2}$   
 F  $2$

8 Simplify  $\left(\frac{a^{\frac{1}{2}}b^{-2}}{a^{-1}b^{\frac{1}{2}}}\right)^4$ .

- A  $a^2b^{-10}$
- B  $a^2b^{-6}$
- C  $a^6b^{10}$
- D  $a^6b^{-6}$
- E  $a^6b^{-10}$
- F  $a^{-2}b^{10}$

9 The number  $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}}$  is equal to

- A 4
- B  $\frac{8}{\sqrt{2}}$
- C  $\sqrt{\frac{2}{34}}$
- D 8
- E  $2\sqrt{2}$
- F  $4\sqrt{2}$

10 Given that  $2^x = 5$  and  $2^y = 20$ , find the value of  $2^{2x-y}$ .

- A  $\frac{1}{4}$
- B  $\frac{5}{4}$
- C  $\frac{5}{2}$
- D  $\frac{1}{10}$
- E  $\frac{25}{2}$
- F 5

11 Solve  $x^{\frac{3}{2}} = 8x^{-\frac{3}{2}}$  for  $x > 0$ .

- A  $x = 2\sqrt{2}$
- B  $x = 16$
- C  $x = 8$
- D  $x = \sqrt{2}$
- E  $x = 2$
- F  $x = 4$

**12** How many real solutions does the equation  $2^{x^2} = 16 \times 2^{-3x}$  have?

**A** 3

**B** 2

**C** 4

**D** 1

**E** 0

**F** infinitely many